



Presenter One-Pager — Know This Cold

Carry this in your pocket. Read it five times on the train to the venue. You should be able to answer every line from memory.

Identity

- **Team name:** SUBARNAREKHA — the river that flows from Jharkhand through Bengal and empties into the Bay of Bengal at Odisha's Balasore coast. Subarna = gold, Rekha = streak. The river carries gold to Odisha. We're infrastructure built *for* Odisha, flowing *from* open-source.
- **Project name:** AaaS — Accessibility as a Service — built for Odisha, designed for every Indian public institution.
- **One-liner:** *One button on any government website that reads, translates, simplifies, scans and listens — in Odia, by voice. Odisha first, India next.*
- **Team:** Pratikshya Padhi (Class XI), the only member · Odisha Adarsha Vidyalaya, Jamdhar · Guide teacher: Somali Priyadarshini Mohanty (98611 37602) · Odiapreneur 3.0, theme *Accessibility, Ecommerce & Cyber Security*.
- **License:** Apache-2.0. Say this early in any conversation.

What it is (30-second version)

One floating ଅ button that works on **any** website — via a one-line embed or our browser extension — backed by four small services running **on-prem** (this laptop today; OCAC's data centre in production):

1. **The widget** — six features behind six tiles: Read aloud (with follow-along highlight), Translate → Odia, Easy Read simplification, Read document (OCR for scanned notices/PDFs with side-by-side view), Speak to fill (guided voice form filling), Voice command navigation. Plus OpenDyslexic "Comfortable letters", a reading ruler, and hover/selection speech.
2. **The AI services** — Odia STT (AI4Bharat IndicWav2Vec), translation (AI4Bharat IndicTrans2), speech (Meta MMS-TTS, or/hi/en), OCR (Tesseract), and a fully offline rule-based text simplifier — behind one gateway with API-key tenancy.

Nothing leaves the machine. Every feature has a deterministic mock fallback, so the demo cannot die.

What is BUILT vs planned (get this right)

Built, working, demoable now:

- All six widget tiles + comfort toggles, verified on real government sites (jajpur.odisha.gov.in, bseodisha.ac.in, odisha.gov.in, ssepd.odisha.gov.in)
- Browser extension (MV3, v0.4.0) — injects the widget into any site
- Four services + gateway, 240+ automated checks green
- Odia number-speech in TTS (digits → spoken Odia words)

- Name **transliteration** (not translation) for voice form fill
- Spoken-number understanding (Odia words, phonetic English, "double" forms)
- Portable package (SUBARNAREKHA-Setup.zip): the full real stack runs offline on any Windows laptop, nothing to install

Roadmap (say "designed for, not built"):

- More Indic languages beyond or/hi/en (the engines already support them)
- Sign-language avatar — a three.js concept demo exists, deliberately deferred
- WCAG audit tooling for site owners

Cut (say "we cut it for focus" if asked):

- The accessible mock-exam module — removed, not deferred.

Numbers to memorise

Odisha (say these first to any Odisha-audience judge)

WHAT	NUMBER
Odia speakers worldwide	~42M (L1: ~37.5M per 2011 Census)
Odisha population	~4.2 crore (2011 Census)
Odisha literacy	72.9% — about 27% aged 7+ cannot read (2011 Census)
Persons with disabilities, all India	2.68 crore = 2.21% (2011 Census)
Odia — declared classical language	2014 (6th in India)
State IT agency	OCAC
State digital mandate	5T + Mo Sarkar

Platform

WHAT	NUMBER
Widget size, self-contained	~520 KB, zero dependencies
Services	4 (gateway, TTS, STT, translate+OCR+simplify)
Automated checks	240+ across 5 suites
Languages live today	Odia, Hindi, English
Real sites verified	6 incl. jajpur.odisha.gov.in, india.gov.in
Cloud calls required	0 (fully on-prem)

The stack (never misname these)

JOB	MODEL / ENGINE	WHO MADE IT
Speech → text (Odia)	IndicWav2Vec	AI4Bharat
Translation	IndicTrans2 (distilled 200M)	AI4Bharat
Text → speech (or/hi/en)	MMS-TTS	Meta
OCR (scans, PDFs)	Tesseract	open source
Simplification (Easy Read)	rule-based, ours	us — fully offline

NOT in the stack: Whisper, IndicConformer, Piper, NLLB (available as an option, not shipped), any LLM, any cloud API. If a judge asks "why not an LLM?" — "Rules are deterministic, auditable, run offline on a ₹30,000 laptop, and can't hallucinate a wrong deadline into a government notice."

Differentiation (the three answers that win)

1. **"Doesn't Bhashini do this?"** — Bhashini is the language engine layer; we're the accessibility product ON TOP of engines like it. Bhashini gives India voices; AaaS gives Odisha's disabled citizens hands, eyes and ears — guided form filling, scanned-notice reading, never-dead-end voice nav. We'd plug Bhashini in as a backend tomorrow.
2. **"Why not just use Chrome's built-in reader?"** — Chrome doesn't speak Odia numbers, doesn't OCR scanned circulars into side-by-side Odia, doesn't fill forms by voice, and doesn't transliterate names. We do those four things *because* we're built Odia-first, not translated-in.
3. **"What's hard about this?"** — The demo hides the hard parts: MMS-TTS is mute on digits (we verbalize numbers in Odia), STT writes English words phonetically in Odia script (we sound-match them to real links), naive translation turns the name Purnnachandra into "full moon" (we transliterate), and government sites hide their most important links in carousels (we reach them anyway).

Honest weaknesses (own them before judges find them)

- Odia OCR quality on poor-quality scans is modest (tessdata_fast) — printed notices yes, handwriting no.
- STT needs reasonably clear speech; booth noise hurts — that's why the guided form fill retries and the voice nav shows "did you mean" buttons.
- Translating a long page for the first time takes a couple of minutes on CPU; after that it is cached and instant.
- One-student team, hackathon timeline — breadth over per-feature depth, by design.